

Collaborative Cache-Aided Relaying Networks: Performance Evaluation and System Optimization

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Abstract—This paper studies a multi-tier cache-aided relaying network, where the destination D is randomly located in the network and it requests files from the source S through the help of cache-aided base station (BS) and N relays. In this system, the multi-tier architecture imposes a significant impact on the system collaborative caching and file delivery, which brings a big challenge to the system performance evaluation and optimization. To address this problem, we first evaluate the system performance by deriving analytical outage probability expression, through fully taking into account the random location of the destination and different file delivery modes related to the file caching status. We then perform the asymptotic analysis on the system outage probability when the signal-to-noise ratio (SNR) is high, to enclose some important and meaningful insights on the network. We further optimize the caching strategies among the relays and BS, to improve the network outage probability. Simulations are performed to show the effectiveness of the derived analytical and asymptotic outage probability for the proposed caching strategy. In particular, the proposed caching is superior to the conventional caching strategies such as the most popular content (MPC) and equal probability caching (EPC) strategies.

Index Terms—Collaborative caching, relay selections, file delivery, outage probability.

I. INTRODUCTION

With the rapid development of multimedia technologies and popular use of mobile applications such as Youtube, Netflix and TikTok, mobile data traffic has increased profoundly in recent years. As estimated by Cisco, the global network traffic will reach to about 4.8 ZB per year by 2025, whereas 71% of data are produced by the wireless and mobile devices [1], which brings huge challenges for the whole network from the perspectives of both communication and computing. To

alleviate the overhead of the core network and reduce the mobile traffic, many communication and computing techniques have been proposed by researchers [2]–[6], among which caching technology is a promising one. The critical idea behind the caching is deploying wireless storage in the edge of the networks and prefetching the popular files to assist the transmission, which can significantly boost the quality of experience (QoE) of users. [7].

A. Literature Review

The key design in the wireless caching is the caching strategy, which determines a part of files pre-stored in the network nodes, as the storage space is generally limited [8]. There are two typical caching strategies in the wireless networks: *most popular contents (MPC)* and *largest content diversity (LCD)* caching strategies. MPC always stores the most popular files in the network nodes, which can achieve the largest cooperation gain among nodes, while LCD stores each file at most one node only, which can obtain the largest caching diversity [9]. From these two caching strategies, several works have been performed to devise a hybrid caching strategy to achieve a proper trade-off between the cooperation and caching gain [10]–[13]. For the wireless stochastic networks, the probabilistic caching should be used to exploit the dynamic characteristics of the network status. In this direction, the authors in [14] achieved an optimal cache placement in wireless networks, whereas the optimal cache probability for each file was obtained by using the convex optimization. After that, the authors in [15] extended this study to device-to-device (D2D) communication and maximized the throughput of the D2D network by deriving the optimal cache probabilities. In addition, a similar extension about probabilistic caching was investigated in [16], where cache probabilities were solved for the files with time-varying popularity. In further, the outage probability of probabilistic caching in cellular networks was also analyzed in [17], [18] to reveal the effect of cell load and hit probability on the system performance.

Besides the caching technique, relaying is another promising technique to support the development of wireless networks, which can enlarge the coverage area and enhance the transmission quality [19]–[22]. Typical relaying protocols including *amplify-and-forward (AF)*, *decode-and-forward (DF)* and *denoise-and-forward (DnF)* have been proposed and widely investigated in the literature, where the system performance were extensively studied such as capacity, outage probability and symbol error rate (SER) [23]–[25]. For such studies, [26] analyzed multiuser massive multiple-input multiple-output (MU-MIMO) relaying networks, by maximizing the achievable

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data rate by power allocation. A novel intelligent reflecting surface (IRS)-assisted relaying network was proposed in [27], which could significantly improve the system performance. In [28], the relaying networks with secrecy constraints were studied for *non-orthogonal multiple access (NOMA)*, where a novel decoding scheme was proposed to ensure secure communications.

Recently, relaying has been incorporated into the cache-aided networks, to further increase the coverage and enhance the transmission performance. In this direction, the authors in [29] considered a multi-relay network with cache and optimized the cache placement through minimizing the outage probability. Moreover, the impact of eavesdroppers on cache-aided relaying networks was analyzed in [12], whereas the caching strategy was designed to guarantee the security of the network. So far, there has been little work on the study of multi-tier cache-aided relaying networks, where the multi-tier architecture imposes a profound impact on the collaborative caching and file delivery [30], which brings a big challenge to the system performance evaluation and optimization. This is the major motivation of this paper.

B. Contribution

This paper studies a collaborative cache-aided relaying network, where the destination D is randomly located in the network and it requests files from the source S through the help of cache-aided base station (BS) and N relays. As the cache size at the relays and BS is limited in practice, only a part of the requested files can be pre-stored according to the probabilistic caching model. For this network, we begin with the first critical question: “How to evaluate the system performance by jointly taking into account the file popularity and transmission modes affected by the file caching status?”. To answer this question, we evaluate the system performance by providing the analytical expression for the outage probability, through incorporating the random location of the destination and different file delivery modes related to the file caching status. We proceed with one more important question: “What are the insights on the network design from the system parameters?”. To tackle this question, we perform the asymptotic expression for outage probability when *signal-to-noise ratio (SNR)* is high, to enclose some important and meaningful insights on the system design. We further consider another important question: “How to optimize the multi-tier cache-aided relaying network by fully utilizing the collaborative caching resources?”. To answer this question, we optimize the caching strategies among the relays and BS to improve the system outage probability by utilizing an improved *block coordinate descent (BCD)* algorithm. Simulations are finally performed to verify the effectiveness of the derived analytical and asymptotic outage probability of the proposed caching strategy. In particular, the proposed caching is superior to conventional caching strategies such as MPC and EPC.

II. SYSTEM MODEL

In this section, we first introduce the network model of collaborative cache-aided relaying networks and the file popularity model. After that, the file delivery ways are described.

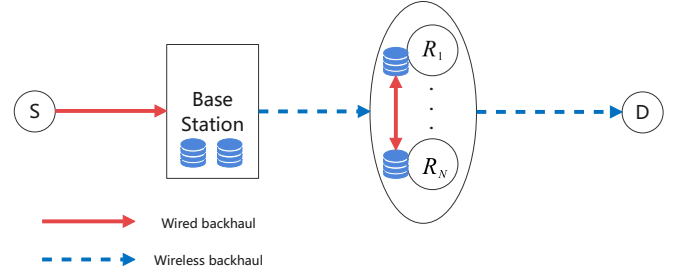


Figure 1. Illustration of cache-aided relaying networks.

Table I
NOTATIONS SUMMARY

Notation	Definition
C_1, C_2	Cache size of relay and BS
K	Number of files
N	Number of relays
P_R, P_{BS}	Transmit power of relay and BS
Q	Number of bits in the file
N_o	Noise power
Ω_k	Relay set that caches the k -th file
a_k	Popularity of the k -th file
b_k	Rank of the k -th file
d_1	Distance between D and relay cluster
d_2	Distance between BS and relay cluster
$g_{1,k}$	Cardinal number of Ω_k
$g_{2,k}$	Indicator of whether the k -th file is cached by the BS
$h_{R_n,D}$	Channel parameter of the R_n - D link
h_{BS,R_n}	Channel parameter of the BS- R_n link
$q_{1,k}, q_{2,k}$	Cache probability of the k -th file
$t_{k,1}, t_{k,2}, t_{k,3}, t_{k,4}$	Delivery latency
η, τ	Plateau and skewness of MZipf distribution
σ	Decay coefficient

Some main symbols used in this paper are summarized in Table 1.

A. Network model

Fig. 1 shows a collaborative cache-aided relaying network, where the destination D requests files from the source S ¹, through the help of cache-aided base station (BS) and N relays. The set of the N relays is denoted as $\mathcal{N} = \{1, 2, \dots, N\}$, and these relays constitute a cluster. The destination D is uniformly located in the coverage of relays and it cannot directly communicate with the source S due to obstacles. To aid the file delivery from the S to the destination D , the relays operate in DF² relaying protocol, and each relay connects to other ones through wired intra-backhaul. Besides, the BS communicates with the source S via a wired core network, which consists of routers and optical fiber. Moreover, a total number of K files will be requested by the D . The BS and relays are equipped with a cache of a finite size, which can prefetch some popular files and help deliver them, relieving the

¹In practice, the source S can be a cloud server.

²We consider the DF relaying protocol in this work, and we can easily extend this work to other relaying protocols such as AF and DnF with some slight modifications. Some existing works also focus on cache placement for the networks with other relaying protocols, which can be found in the literature such as [29], [31].

overhead of the core networks. The cache size of the relays and BS is C_1 and C_2 , respectively, where $C_1 < C_2 < K$ holds in practical wireless networks. In other words, the BS and relays can only cache a part of the files.

As to the wireless channels in the system, the links between the relays and BS or destination are assumed to be subject to Rayleigh fading. We adopt the typical path-loss model to characterize Rayleigh fading with a decay coefficient σ . Let d_1 and d_2 denote the distances from the relays to the destination and BS, respectively. In this work, we regard d_2 as a constant, which is reasonable when the distance between the relaying cluster and BS is fixed in practical networks. Then, we use $h_{R_n,D} \sim \mathcal{CN}(0, d_1^{-\sigma})$ to denote the channel parameter of the R_n -D link, and we use $h_{BS,R_n} \sim \mathcal{CN}(0, d_2^{-\sigma})$ to represent that of the BS- R_n link. For simplicity, we set the associated instantaneous channel gain to $|h_{R_n,D}|^2 = d_1^{-\sigma} v_{1n}$ and $|h_{BS,R_n}|^2 = d_2^{-\sigma} v_{2n}$, where v_{1n} and v_{2n} are two exponentially distributed random variables with unity mean.

B. File popularity and caching strategy

The file popularity characterizes the popular extent of the files accessed by users in the network, and it can be estimated by analyzing historical user behaviour. In this work, we use the MZipf distribution [32], a more general statistical distribution, to model the file popularity, instead of the Zipf distribution. Accordingly, the popularity of the k -th file is

$$a_k = \frac{(b_k + \tau)^{-\eta}}{\sum_{k=1}^K (b_k + \tau)^{-\eta}}, \quad (1)$$

where b_k denotes the rank of file k , and $\tau \geq 0$ and $\eta > 0$ represent the parameters of plateau and skewness, respectively. Note that τ determines the plateau shape near the distribution's left-most part. The larger the value of τ , the more flattened the head will be. In particular, when $\tau = 0$, the MZipf degrades to the conventional Zipf distribution. Besides, η is the skewness factor that decides the concentration of file popularity. A larger value of η indicates a higher concentrated file popularity, and a smaller one refers to uniform-like popularity. Based on the file popularity, we can make some caching decisions according to the probabilistic caching model. Let $q_{1,k}$ and $q_{2,k}$ denote the caching probability of the k -th file in the relays and BS, respectively, where the following constraints on the caching probability should be satisfied,

$$\sum_{k=1}^K q_{1,k} \leq C_1, \quad (2)$$

$$\sum_{k=1}^K q_{2,k} \leq C_2. \quad (3)$$

C. File delivery

In practice, file delivery is affected by the caching strategy, as the caching strategy directly determines the storage position of the requested files. Let Ω_k denote the relay set, in which the k -th file has been cached, and $g_{1,k} = |\Omega_k|$ denotes the cardinal number. In addition, we use $g_{2,k} = 1$ to indicate that the BS has stored the k -th file or not when $g_{2,k} = 0$. The

delivery mode of the k -th file depends on the specific values of $g_{1,k}$ and $g_{2,k}$, detailed as follows,

1) *Mode 1 with $g_{1,k} \geq 1$* : In this situation, the k -th file has been stored in at least one relay, and hence it can be directly fetched by D from these relays. The best relay $R_{n_1^*}$ can be selected among the $g_{1,k}$ relays in Ω_k ³, through maximizing the received SNR at the destination,

$$n_1^* = \arg \max_{n \in \Omega_k} |h_{R_n,D}|^2, \quad (4)$$

$$= \arg \max_{n \in \Omega_k} v_{1n}. \quad (5)$$

Then, the latency of delivering the k -th file is

$$t_{k,1} = \frac{Q}{\log_2 \left(1 + P_R d_1^{-\sigma} v_{1n_1^*} / N_o \right)}, \quad (6)$$

where Q is the number of bits in the file, P_R is the transmit power at the relays, and N_o is the variance of *additive Gaussian white noise (AWGN)* at the destination [33].

Note that the above file delivery utilizes $g_{1,k}$ relays only, while the residual $(N - g_{1,k})$ relays are not used. When the channels associated with the $g_{1,k}$ relays are in poor state, the delivery latency $t_{k,1}$ will become large. Hence, we should also exploit the residual $(N - g_{1,k})$ relays for the file delivery. Among these relays, we can select another best relay $R_{n_2^*}$ to assist the file delivery as

$$n_2^* = \arg \max_{n \in \mathcal{N} \setminus \Omega_k} |h_{R_n,D}|^2, \quad (7)$$

$$= \arg \max_{n \in \mathcal{N} \setminus \Omega_k} v_{1n}. \quad (8)$$

In this case, the delivery latency becomes

$$t_{k,2} = \frac{Q}{\log_2 \left(1 + P_R d_1^{-\sigma} v_{1n_2^*} / N_o \right)} + T_1, \quad (9)$$

where T_1 denotes a constant transmission latency in the wired intra-backhaul link.

By jointly taking into account the relays in the hit group and the relays in the missed group, we can obtain the best relay R_{n^*} by comparing the delivery latency $t_{k,1}$ and $t_{k,2}$,

$$n^* = \arg \min_{n \in \{n_1^*, n_2^*\}} (t_{k,1}, t_{k,2}). \quad (10)$$

In this way, all the relays are exploited to assist the file delivery, regardless of caching the file or not.

2) *Mode 2 with $g_{1,k} = 0$ and $g_{2,k} = 1$* : When the k -th file has not been cached in any relay, while the BS has cached, D can fetch the file from the BS with the help of N DF relays. In this case, the dual-hop DF relaying should be utilized, and

³To implement this relay selection, D firstly broadcasts the file request to all relays, and then those relays that have stored the requested file will send pilot signals to D . Therefore, D can utilize the received pilot signals to estimate the channel parameters and perform the relay selection according to (4).

we can perform the conventional max-min criterion to select the best relay R_{n^*} ,

$$n^* = \arg \max_{n \in \mathcal{N}} \min \left(P_{BS} |h_{BS,R_n}|^2, P_R |h_{R_n,D}|^2 \right), \quad (11)$$

$$= \arg \max_{n \in \mathcal{N}} \min \left(P_{BS} d_2^{-\sigma} v_{2n}, P_R d_1^{-\sigma} v_{1n} \right), \quad (12)$$

where P_{BS} is the transmit power at the BS. In this case, the delivery latency is

$$t_{k,3} = \frac{Q}{\frac{1}{2} \log_2 \left(1 + \tilde{P}_{n^*}/N_o \right)}, \quad (13)$$

where $\frac{1}{2}$ comes from the two-phase data transmission, and notation $\tilde{P}_n$ is denoted by

$$\tilde{P}_n = \min \left(P_{BS} d_2^{-\sigma} v_{2n}, P_R d_1^{-\sigma} v_{1n} \right). \quad (14)$$

3) *Mode 3 with $g_{1,k} = 0$ and $g_{2,k} = 0$* : In this case, neither the BS nor relays has stored the requested file, and hence D has to fetch it from the S . Accordingly, the delivery latency is composed of the latency from the BS to D and that from S to the BS, given by

$$t_{k,4} = \frac{Q}{\frac{1}{2} \log_2 \left(1 + \tilde{P}_{n^*}/N_o \right)} + T_2, \quad (15)$$

where T_2 is a constant transmission latency in the wired backhaul between the BS and the source S , and the best relay R_{n^*} is chosen according to the criterion in (12).

By summarizing the above three file delivery modes, we can write the latency t_k of delivering the k -th file as,

$$t_k = \begin{cases} \min(t_{k,1}, t_{k,2}), & \text{If } g_{1,k} \geq 1 \\ t_{k,3}, & \text{If } g_{1,k} = 0, g_{2,k} = 1 \\ t_{k,4}, & \text{If } g_{1,k} = 0, g_{2,k} = 0 \end{cases} \quad (16)$$

From (16), we can find that the delivery mode has a significant impact on the system performance. In particular, the usage of wireless caching can help reduce the transmission latency on average. In the following, we will provide the system outage probability analysis for the considered collaborative cache-aided relaying networks, and give meaningful insights on the system design and optimization.

III. OUTAGE PROBABILITY ANALYSIS

This part presents the system performance evaluation of the cache-aided relaying networks with K files in total. Specifically, we first present the analytical form of the outage probability, and then we derive the asymptotic expression with high SNR to enclose the insight for system design.

A. Analytical Outage Probability

For the delivery of the k -th file, the outage event occurs if the delivery latency t_k exceeds a predetermined threshold γ_{th} .

Hence, the system outage probability of delivering the total K files is given by

$$P_{out} = \sum_{k=1}^K a_k \Pr(t_k > \gamma_{th}). \quad (17)$$

By incorporating the three delivery modes associated with different values of $g_{1,k}$ and $g_{2,k}$, we can further specify P_{out} as

$$P_{out} = \sum_{k=1}^K a_k \left(\sum_{g_{1,k}=1}^N \Pr(g_{1,k}) \cdot P_{out,1}(g_{1,k}) + \Pr(g_{1,k} = 0, g_{2,k} = 1) \cdot P_{out,2} + \Pr(g_{1,k} = 0, g_{2,k} = 0) \cdot P_{out,3} \right), \quad (18)$$

where $\Pr(g_{1,k})$, $\Pr(g_{1,k} = 0, g_{2,k} = 1)$ and $\Pr(g_{1,k} = 0, g_{2,k} = 0)$ return the probability of the specific $g_{1,k}$ and $g_{2,k}$, while $P_{out,1}(g_{1,k})$, $P_{out,2}$ and $P_{out,3}$ correspond to the outage probability of the delivery mode 1, 2, and 3, respectively.

From the theory of probability, we can calculate $\Pr(g_{1,k})$, $\Pr(g_{1,k} = 0, g_{2,k} = 1)$ and $\Pr(g_{1,k} = 0, g_{2,k} = 0)$ as

$$\Pr(g_{1,k}) = \binom{N}{g_{1,k}} q_{1,k}^{g_{1,k}} (1 - q_{1,k})^{N - g_{1,k}}, \quad (19)$$

$$\Pr(g_{1,k} = 0, g_{2,k} = 1) = (1 - q_{1,k})^N q_{2,k}, \quad (20)$$

$$\Pr(g_{1,k} = 0, g_{2,k} = 0) = (1 - q_{1,k})^N (1 - q_{2,k}). \quad (21)$$

We then calculate the analytical outage probabilities $P_{out,1}(g_{1,k})$, $P_{out,2}$ and $P_{out,3}$ for the three delivery modes, which are provided in the following three theorems,

Theorem 1. The analytical expression of $P_{out,1}(g_{1,k})$ is given by

$$P_{out,1}(g_{1,k}) = 1 + \frac{2}{R^2} \sum_{n_1=0}^{g_{1,k}} \sum_{n_2=1}^{N - g_{1,k}} \binom{g_{1,k}}{n_1} \binom{N - g_{1,k}}{n_2} \times (-1)^{(n_1+n_2)} \frac{\gamma\left(\frac{2}{\sigma}, \beta_{1,1} R^\sigma\right)}{\sigma \beta_{1,1}^{\frac{2}{\sigma}}} + \frac{2}{R^2} \sum_{n_1=1}^{g_{1,k}} \binom{g_{1,k}}{n_1} (-1)^{n_1} \frac{\gamma\left(\frac{2}{\sigma}, \beta_{1,2} R^\sigma\right)}{\sigma \beta_{1,2}^{\frac{2}{\sigma}}}, \quad (22)$$

where R is the maximum coverage of the relays, $\gamma(\cdot, \cdot)$ denotes the lower incomplete Gamma function, and $\gamma_{1,1}$, $\gamma_{1,2}$, $\beta_{1,1}$, and $\beta_{1,2}$ are given by

$$\begin{cases} \gamma_{1,1} = 2^{\frac{Q}{\gamma_{th}}} - 1, \\ \gamma_{1,2} = 2^{\frac{Q}{\gamma_{th} - T_1}} - 1, \end{cases}, \quad \begin{cases} \beta_{1,1} = \frac{(n_1 \gamma_{1,1} + n_2 \gamma_{1,2}) N_o}{P_R}, \\ \beta_{1,2} = \frac{(n_1 \gamma_{1,1}) N_o}{P_R}. \end{cases} \quad (23)$$

Proof. See Appendix A. ■

Theorem 2. The analytical expression of $P_{out,2}$ is given by

$$P_{out,2} = 1 + \frac{2}{R^2} \sum_{n=1}^N \binom{N}{n} (-1)^n \times \exp\left(-\frac{n \gamma_2 N_o}{P_{BS} d_2^{-\sigma}}\right) \frac{\gamma\left(\frac{2}{\sigma}, \beta_2 R^\sigma\right)}{\sigma \beta_2^{\frac{2}{\sigma}}}, \quad (24)$$

with

$$\begin{cases} \gamma_2 = 2^{\frac{2Q}{\gamma_{th}}} - 1, \\ \beta_2 = \frac{n\gamma_2 N_o}{P_R}. \end{cases} \quad (25)$$

Proof. See Appendix B. ■

Theorem 3. The analytical expression of $P_{out,3}$ is given by

$$P_{out,3} = 1 + \frac{2}{R^2} \sum_{n=1}^N \binom{N}{n} (-1)^n \times \exp\left(-\frac{n\gamma_3 N_o}{P_{BS} d_2^{-\sigma}}\right) \frac{\gamma\left(\frac{2}{\sigma}, \beta_3 R^\sigma\right)}{\sigma \beta_3^{\frac{2}{\sigma}}}, \quad (26)$$

with $\gamma_3 = 2^{\frac{2Q}{\gamma_{th} - T_2}} - 1$ and $\beta_3 = \frac{n\gamma_3 N_o}{P_R}$.

Proof. The proof of Theorem 3 is similar to that of Theorem 2, which is detailed in Appendix B. ■

By incorporating Theorem 1, 2 and 3 into (18), the analytical expression of the system outage probability P_{out} is finally obtained. Note that this expression can be readily computed and the system outage probability can be easily evaluated in the whole range of SNR.

B. Asymptotic Outage Probability

To obtain some important and meaningful instructions on the network design and optimization, we further perform the asymptotic analysis with large SNR. In this case, we can use the approximation of $e^x \simeq 1 + x$ for a small $|x|$, and then approximate expressions of $P_{out,1}(g_{1,k})$, $P_{out,2}$ and $P_{out,3}$, in the following,

Theorem 4. The asymptotic expressions of $P_{out,1}(g_{1,k})$, $P_{out,2}$ and $P_{out,3}$ with large SNR are respectively given by,

$$P_{out,1}(g_{1,k}) \simeq \frac{2\gamma_{1,1}^{g_{1,k}} \gamma_{1,2}^{N-g_{1,k}} R^{N\sigma} N_o^N}{P_R^N (\sigma N + 2)}, \quad (27)$$

$$P_{out,2} \simeq 2\gamma_2^N N_o^N \sum_{n=0}^N \binom{N}{n} P_R^{n-N} P_{BS}^{-n} d_2^{\sigma n} \frac{R^\delta}{\delta + 2}, \quad (28)$$

$$P_{out,3} \simeq 2\gamma_3^N N_o^N \sum_{n=0}^N \binom{N}{n} P_R^{n-N} P_{BS}^{-n} d_2^{\sigma n} \frac{R^\delta}{\delta + 2}, \quad (29)$$

where $\delta = (N - n)\sigma$.

Proof. See Appendix C. ■

By incorporating Theorem 4 into (18), we can obtain the asymptotic outage probability P_{out} , and therefore, some important and meaningful instructions on the network design and optimization can be obtained,

- The system diversity order is N , which can indicate that the system can fully exploit the N relays, for all three delivery modes. Moreover, the system outage performance can be rapidly enhanced with a larger number of relays.
- The outage probability of delivering the k -th file decreases with a larger $g_{1,k}$ and $g_{2,k}$, indicating that caching can be used to reduce the file delivery latency. Accordingly, increasing the caching size C_1 and C_2 can

help improve the system whole outage performance by reducing the delivery latency.

- As the outage performance of delivering the k -th file is significantly affected by the associated caching strategy, we should optimize the collaborative caching among the K files, by jointly taking into account both the transmission performance and file popularity.

IV. CACHE STRATEGY OPTIMIZATION

In this part, we investigate how to optimize the caching strategy for the considered collaborative cache-aided relaying network, as it has a significant impact on the values of $g_{1,k}$ and $g_{2,k}$ and hence affects the system outage performance. We formulate the system optimization as

$$\mathbf{P1}: \min_{\{q_{1,k}, q_{2,k} | 1 \leq k \leq K\}} P_{out} \quad (30a)$$

$$\text{s.t.} \quad q_{1,k} \in [0, 1], \quad (30b)$$

$$q_{2,k} \in [0, 1], \quad (30c)$$

$$\sum_{k=1}^K q_{1,k} \leq C_1, \quad (30d)$$

$$\sum_{k=1}^K q_{2,k} \leq C_2, \quad (30e)$$

where (30b) and (30c) indicate the probability constraint, while (30d) and (30e) are the constraint on the caching size at the relays and BS, respectively. To tackle such optimization, we firstly analyze the characteristic of **P1**, whose non-convex property is given by the following lemma,

Lemma 1. The problem **P1** is a non-convex optimization problem.

Proof. See Appendix D. ■

From Lemma 1, we can find that it is arduous to obtain the optimal solution to this problem. To tackle this issue, we employ an iterative algorithm based on BCD to give a sub-optimal solution. Specifically, we firstly decompose **P1** into two sub-problems, in which the first sub-problem **P1-1** is to optimize the caching probability at the relays $\{q_{1,k} | 1 \leq k \leq K\}$, with the given caching probability at the BS $\{q_{2,k} | 1 \leq k \leq K\}$. Then, the second sub-problem **P1-2** is to optimize the caching probability at the BS $\{q_{2,k} | 1 \leq k \leq K\}$, after obtaining the caching probability at the relays $\{q_{1,k} | 1 \leq k \leq K\}$. By repeating these two steps, $\{q_{1,k} | 1 \leq k \leq K\}$ and $\{q_{2,k} | 1 \leq k \leq K\}$ can be optimized rapidly. The detailed optimization of the two sub-problems is given as follows,

A. Caching optimization at the relays

With the given $\{q_{2,k} | 1 \leq k \leq K\}$, the problem **P1** can be transformed into **P1-1**, given by

$$\mathbf{P1-1}: \min_{\{q_{1,k} | 1 \leq k \leq K\}} P_{out} \quad (31a)$$

$$\text{s.t.} \quad q_{1,k} \in [0, 1], \quad (31b)$$

$$\sum_{k=1}^K q_{1,k} \leq C_1. \quad (31c)$$

By analyzing the above problem **P1-1**, we can find that it is a convex function, given by

Lemma 2. P_{out} is a convex function with respect to $q_{1,k}$, and it decreases with $q_{1,k}$.

Proof. See Appendix E. ■

From Lemma 2, we further analyze the optimal $q_{1,k}^*$ for the optimization problem **P1-1**,

Lemma 3. The optimal $q_{1,k}^*$ holds

$$\sum_{k=1}^K q_{1,k}^* = C_1. \quad (32)$$

Proof. Let $\sum_{k=1}^K q_{1,k}^* = C_1$, and $\sum_{k=1}^K q_{1,k}^* = C_1 - \epsilon$, where $\epsilon > 0$. From Lemma 2, the outage probability with $\sum_{k=1}^K q_{1,k}^* = C_1$ is smaller than that with $\sum_{k=1}^K q_{1,k}^* = C_1 - \epsilon$. Hence, (32) is proven. ■

As **P1-1** is a convex optimization problem, we can utilize the classical *Karush-Kuhn-Tucker (KKT)* method [34] to solve the optimal caching probabilities $\{q_{1,k}^* | 1 \leq k \leq K\}$, with the given caching probabilities $\{q_{2,k}^* | 1 \leq k \leq K\}$ at the BS. Specifically, a Lagrangian function associated with **P1-1** is firstly constituted as

$$\begin{aligned} \mathcal{L}(q_{1,k}, \lambda, u_k, v_k) = & P_{out} + \lambda \left(\sum_{k=1}^K q_{1,k} - C_1 \right) \\ & + \sum_{k=1}^K \left(u_k (q_{1,k} - 1) - v_k q_{1,k} \right), \end{aligned} \quad (33)$$

where λ is a Lagrangian multiplier on the equality constraint, while u_k and v_k are the Lagrangian multipliers on the inequality constraint. Then, according to the KKT conditions, the following constraints should be satisfied for the optimal $q_{1,k}^*$,

$$\frac{\partial \mathcal{L}}{\partial q_{1,k}^*} = 0, \quad (34)$$

$$\sum_{k=1}^K q_{1,k}^* = C_1, \quad (35)$$

$$q_{1,k}^* \in [0, 1], \quad (36)$$

$$u_k^* \geq 0, \quad (37)$$

$$v_k^* \geq 0, \quad (38)$$

$$u_k^* (q_{1,k}^* - 1) = 0, \quad (39)$$

$$v_k^* q_{1,k}^* = 0. \quad (40)$$

By solving the above equations, we can obtain the conditions of the optimal $\{q_{1,k}^* | 1 \leq k \leq K\}$ for the problem **P1-1**, as follows,

Theorem 5. Let

$$A_k = q_{2,k} P_{out,2} + (1 - q_{2,k}) P_{out,3}, \quad (41)$$

and define the single-file outage function

$$G_k(q) = \sum_{g=1}^N \binom{N}{g} P_{out,1}(g) q^g (1-q)^{N-g} + A_k (1-q)^N. \quad (42)$$

The optimal $q_{1,k}^*$ for the problem **P1-1** and the associated Lagrangian multiplier λ^* should satisfy

$$q_{1,k}^* = \begin{cases} 0, & \text{If } a_k G'_k(0) + \lambda^* \geq 0, \\ 1, & \text{If } a_k G'_k(1) + \lambda^* \leq 0, \\ q_{1,k}(\lambda^*), & \text{Otherwise,} \end{cases} \quad (43)$$

where $q_{1,k}(\lambda^*) \in (0, 1)$ is the solution of

$$a_k G'_k(q_{1,k}(\lambda^*)) + \lambda^* = 0, \quad (44)$$

with the interior derivative

$$\begin{aligned} G'_k(q) = & \sum_{g=1}^N \binom{N}{g} P_{out,1}(g) q^{g-1} (1-q)^{N-g-1} (g - Nq) \\ & - N A_k (1-q)^{N-1}. \end{aligned} \quad (45)$$

The endpoint values $G'_k(0)$ and $G'_k(1)$ are evaluated by continuous limits, equivalently by the Bernstein derivative form in Appendix E.

Proof. See Appendix F. ■

From Theorem 5, $q_{1,k}^*$ can be obtained by a one-dimensional search for a given λ , and λ^* can be further adjusted to satisfy $\sum_{k=1}^K q_{1,k}^* = C_1$. Accordingly, the efficient bisection search can be employed to find the optimal $\{q_{1,k}^* | 1 \leq k \leq K\}$ and λ^* , as shown in Algorithm 1. Specifically, the search starts from the midpoint of an initial range $[\lambda_{min}, \lambda_{max}]$. According to (43), we can derive the value $q_{1,k}^*$ for the current λ . Then we check whether the constraint in (32) is satisfied or not. **If it is satisfied within a prescribed tolerance, the search will stop and output the solution.** Otherwise, the search range will be adjusted according to the relationship between $\sum_{k=1}^K q_{1,k}$ and C_1 . Specifically, if $\sum_{k=1}^K q_{1,k} > C_1$, the lower bound will be replaced by the midpoint or the upper bound will be replaced by the midpoint otherwise. Such adjustment will be repeated until the constraint in (32) is satisfied.

B. Caching optimization at the BS

After we obtain the optimal $\{q_{1,k}^* | 1 \leq k \leq K\}$, the system outage probability P_{out} can be simplified as

$$P_{out} = \sum_{k=1}^K \left(B_k + J_k q_{2,k} \right), \quad (46)$$

where

$$\begin{cases} B_k = a_k \left(\sum_{g_{1,k}=1}^N \binom{N}{g_{1,k}} q_{1,k}^{g_{1,k}} (1 - q_{1,k}^*)^{N-g_{1,k}} \right. \\ \quad \times P_{out,1}(g_{1,k}) + (1 - q_{1,k}^*)^N P_{out,3} \Big), \\ J_k = a_k (P_{out,2} - P_{out,3}) (1 - q_{1,k}^*)^N. \end{cases} \quad (47)$$

Algorithm 1: Bisection search of optimal $q_{1,k}^*$ and λ^* for **P1-1**.

```

1 Input  $\lambda_{max}, \lambda_{min}$ ;
2  $lower = \lambda_{min}, upper = \lambda_{max}$ ;
3 while  $lower < upper$  do
4    $\lambda = (lower + upper)/2$ ;
5   if  $a_k G'_k(0) + \lambda \geq 0$  then
6      $q_{1,k}^* = 0$ ;
7   else if  $a_k G'_k(1) + \lambda \leq 0$  then
8      $q_{1,k}^* = 1$ ;
9   else
10    Solve  $a_k G'_k(q_{1,k}) + \lambda = 0$  over  $(0, 1)$  to obtain
11     $q_{1,k}^*$ ;
12  end
13  if  $\sum_{k=1}^K q_{1,k}^* > C_1$  then
14     $lower = \lambda$ 
15  else if  $\sum_{k=1}^K q_{1,k}^* < C_1$  then
16     $upper = \lambda$ ;
17  else if  $\sum_{k=1}^K q_{1,k}^* = C_1$  then
18    break;
19 end
20  $\lambda^* = \lambda$ ;
21 return  $q_{1,k}^*, \lambda^*$ 

```

As $J_k < 0$ holds due to $P_{out,2} < P_{out,3}$, we can find that P_{out} is decreasing with a larger $q_{2,k}$, and hence the optimal solution of $\{q_{2,k}^* | 1 \leq k \leq K\}$ becomes,

$$\sum_{k=1}^K q_{2,k}^* = C_2. \quad (48)$$

Then, the caching optimization at the BS is given by

$$\mathbf{P1-2:} \quad \min_{\{q_{2,k} | 1 \leq k \leq K\}} P_{out} \quad (49a)$$

$$\text{s.t.} \quad q_{2,k} \in [0, 1], \quad (49b)$$

$$\sum_{k=1}^K q_{2,k} \leq C_2, \quad (49c)$$

which is a linear programming problem, and can be solved easily. Specifically, let $j(k)$ denote the rank of J_k in the ascending order, and hence the optimal solution of the problem **P1-2** is

$$q_{2,k}^* = \begin{cases} 1, & \text{If } j(k) \leq C_2, \\ 0, & \text{Else} \end{cases}. \quad (50)$$

Note that the solution of **P1-2** is dispersed, and the rank of J_k is not sensitive to the change of $q_{1,k}$. These situations may cause that the solution of **P1-1** and **P1-2** remain the same in the iterative optimization process, which blocks the convergence of BCD. To address this issue, we soften the result of (50) in each BCD iteration as,

$$\text{SOFTEN}(q_{2,k}^*) = \begin{cases} 1 - r, & \text{If } q_{2,k}^* = 1, \\ r, & \text{If } q_{2,k}^* = 0, \end{cases} \quad (51)$$

Algorithm 2: The proposed soft-BCD based caching.

```

1 Input  $q_{1,k}^*$ ;
2 while stopping criterion is not satisfied do
3   Update  $q_{2,k}^*$  for the given  $q_{1,k}^*$  according to (50);
4   Soften  $q_{2,k}^*$  as described in (51);
5   Solve  $q_{1,k}^*$  for the given softened  $q_{2,k}^*$  by
6     Algorithm 1;
7 end
8 Quantize  $q_{2,k}^*$  to  $\{0, 1\}$ ;
9 return  $q_{1,k}^*, q_{2,k}^*$ 

```

where $r > 0$ is a small disturbance to the solution of **P1-2**, which is helpful to jump over the stagnation. Hence, it is unsuitable for setting the value too big or too small. For instance, a too large r may make the convergence fail, while a too small r may fail to improve the situation. The softened $q_{2,k}^*$ is used only as a heuristic perturbation for updating $q_{1,k}^*$ and does not need to satisfy the BS cache-size constraint during this intermediate step. The final BS caching vector is projected back to the feasible binary set according to (50).

Overall, the proposed soften-BCD caching is summarized in Algorithm 2. In detail, in the first step, we initialize the value of $q_{1,k}^*$ and regard it as the input of the algorithm. Then, we optimize $q_{2,k}^*$ with the given $q_{1,k}^*$ (Step 3). The soft operation follows up (Step 4), and we further use Algorithm 1 to update $q_{1,k}^*$ (Step 5). Steps 3-5 will be repeated until the stopping criterion is satisfied⁴. Finally, we use a round operation to quantize $q_{2,k}^*$ to $\{0, 1\}$ and output the results of $q_{1,k}^*$ and $q_{2,k}^*$.

V. SIMULATION RESULTS AND DISCUSSIONS

In this part, we conduct simulations and provide numerical results to verify the proposed studies. We will first introduce the general simulation settings, and then present the competitive caching strategies for performance comparison. After that, we demonstrate the effectiveness of the derived analytical and asymptotic expressions for the system outage probability, and further verify the advantages of the proposed caching strategy over the competitive ones.

A. Simulation settings

In this part, we present some common parameter settings for the simulations. Specifically, if not specified, the transmit power at the relays and BS is set to 40dBm, while the noise variance N_o is 10^{-6} . The maximal coverage of the relays is 100m, while the distance between the relays and BS is set to 75m. The wireless links are subject to Rayleigh fading, where we adopt the path-loss model with the decay exponent of 3. Moreover, there are totally 50 files, and the file size Q is normalized to unity. The Mzipf parameters η and τ are set to 1.5 and 0, respectively. In further, the cache size C_1 and

⁴Some classical stopping criteria can be used in this paper. For example, the iteration can be stopped when the obtained solution almost remains unchanged. We can also set a target number of iterations, where the algorithm will stop when it reaches the given number.

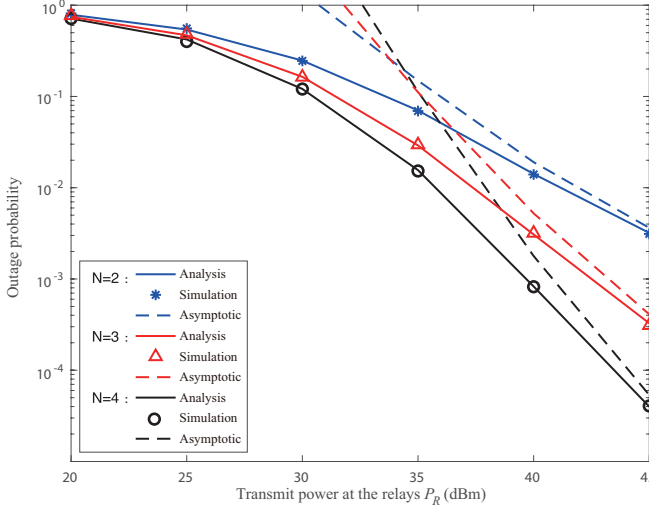


Figure 2. Outage probability of the proposed caching strategy versus the transmit power at the relay P_R , where the transmit power at the BS P_{BS} is 45dBm.

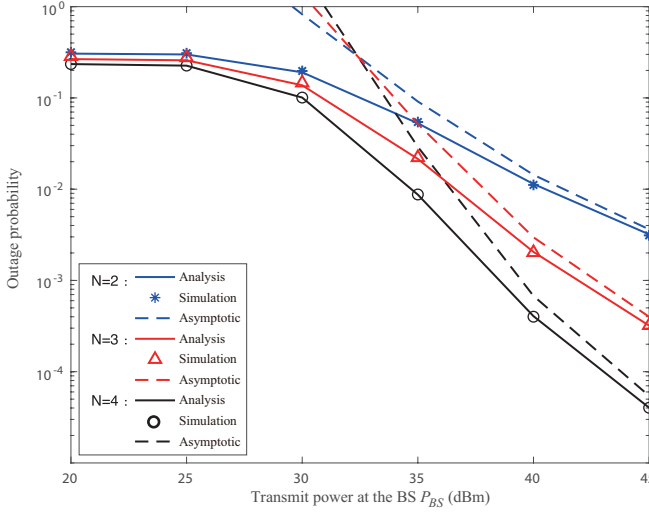


Figure 3. Outage probability of the proposed caching strategy versus the transmit power at the BS P_{BS} , where the transmit power at relay P_R is 45dBm.

C_2 are set to 3 and 10, respectively. The wired transmission latency T_1 and T_2 are set to 0.1s and 0.2s, respectively, and the target latency threshold γ_{th} is set to 1s. In addition, we set r from 0.1 to 0.3 to achieve a better convergence performance in the soften operation.

B. Competing caching strategies

In order to verify the effectiveness of the proposed caching strategy, we provide the following competitive caching strategies for comparison,

- **MPC**: The classical popularity-based caching strategy, where each node in the network only pre-stores the most popular files. It is widely used in the cache-aided relaying network, due to the achieved best signal cooperation gain.
- **EPC**: The equal probabilistic caching strategy, where the relays cache the first $N \times C_1$ files with an equal

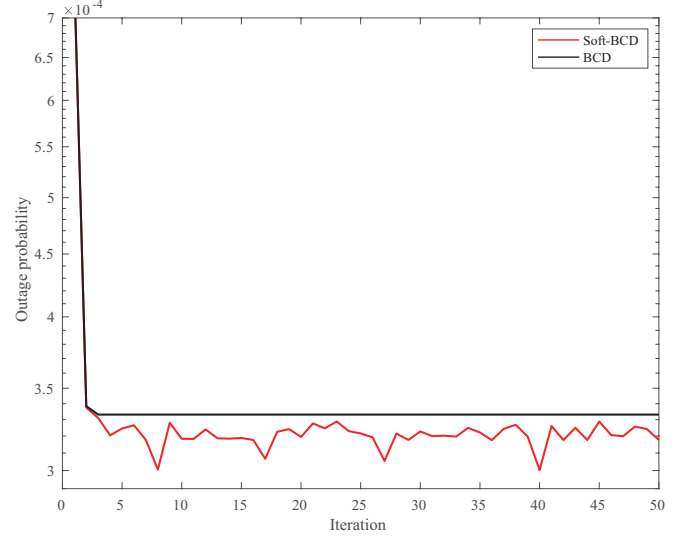


Figure 4. Convergence comparison between the soft-BCD and the conventional BCD, where the transmit power at the relays and BS are 45dBm.

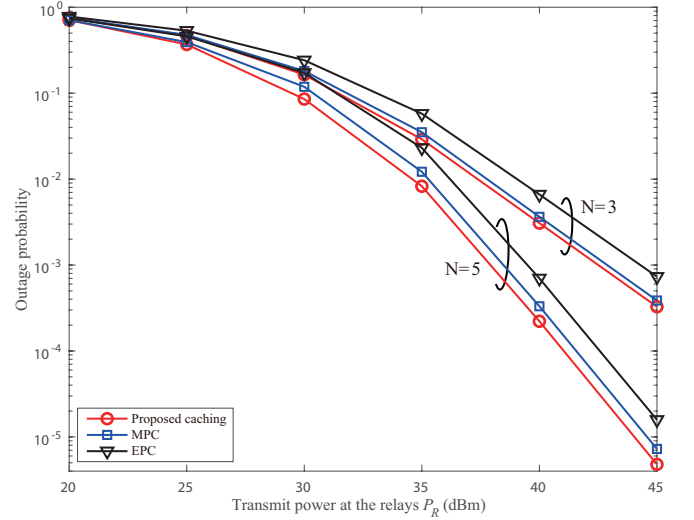


Figure 5. Outage probability of the three caching strategies versus the transmit power at the relays P_R , where the transmit power at BS is 45dBm.

probability. Moreover, the BS caches the residual files with an equal probability. In this way, the largest content diversity is achieved on average.

C. Analysis verification

Fig. 2 depicts the analytical and asymptotic outage probability results of the proposed caching strategy versus the transmit power at the relays, where the transmit power at the BS is 45dBm and the number of relays varies from 2 to 4. From this figure, we can find that the analytical outage probability matches well with the simulation in the entire region of P_R , which verifies the derived analytical expression for the proposed caching. Moreover, the asymptotic outage probability converges to the simulation one, when the value of P_R is large, which validates the effectiveness of the derived asymptotic expression of the outage probability for

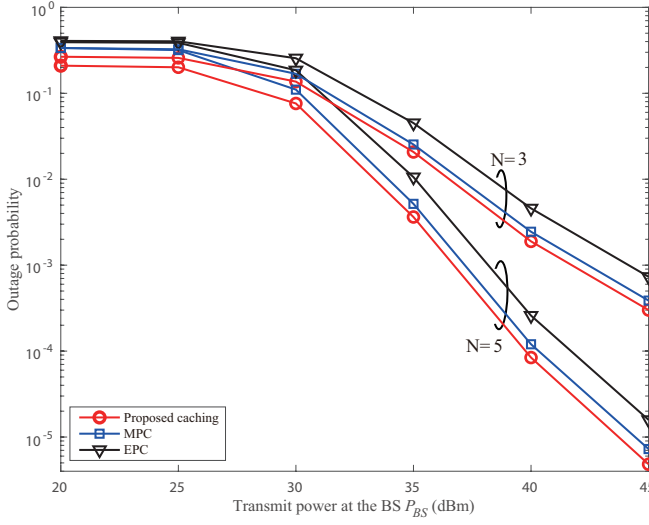


Figure 6. Outage probability of the three caching strategies versus the transmit power at the BS P_{BS} , where the transmit power at relay is 45dBm.

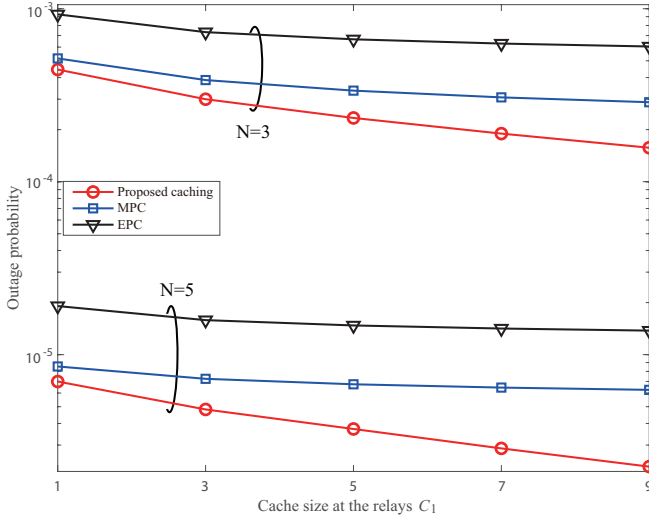


Figure 7. Outage probability of the three caching strategies versus the cache size at the relays C_1 , where the transmit power at relays and BS are 45dBm.

the proposed caching. Moreover, the outage probability of the proposed caching improves with a larger value of P_R , as an increased transmit power at the relays can help reduce the file delivery latency efficiently. The outage probability also improves with a larger value of N , as more relays can provide more spatial diversity, which can be used to enhance the transmission quality. Particularly, the curve slopes of the asymptotic performance get raising with a large N , suggesting that the system diversity order is equal to the number of relays.

Fig. 3 shows the outage probability of the proposed caching versus the transmit power at BS P_{BS} , where $N \in \{2, 3, 4\}$, the transmit power at relays P_R is 45dBm, and P_{BS} varies from 20dBm to 45dBm. From this figure, we can find that the analytical results fit the simulation, and the asymptotic results become convergent to the simulation with a large P_{BS} , which confirms the accuracy of the derived analytical and asymptotic P_{out} for the proposed caching. Moreover, the curve slopes

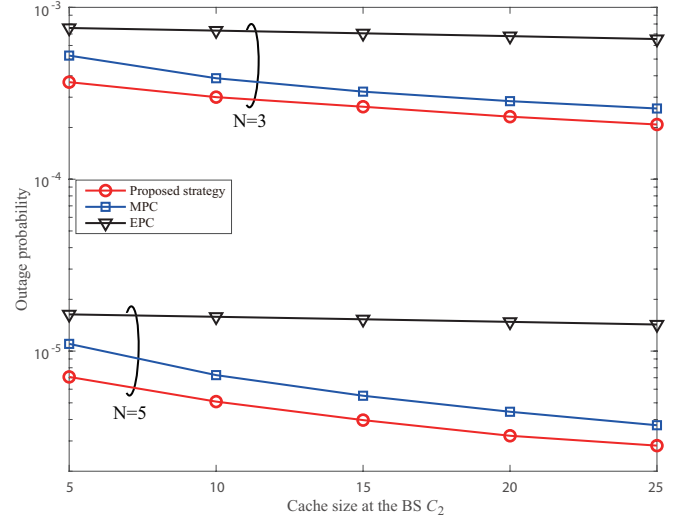


Figure 8. Outage probability of the three caching strategies versus the cache size at the BS C_2 , where the transmit power at the relays and BS are 45dBm.

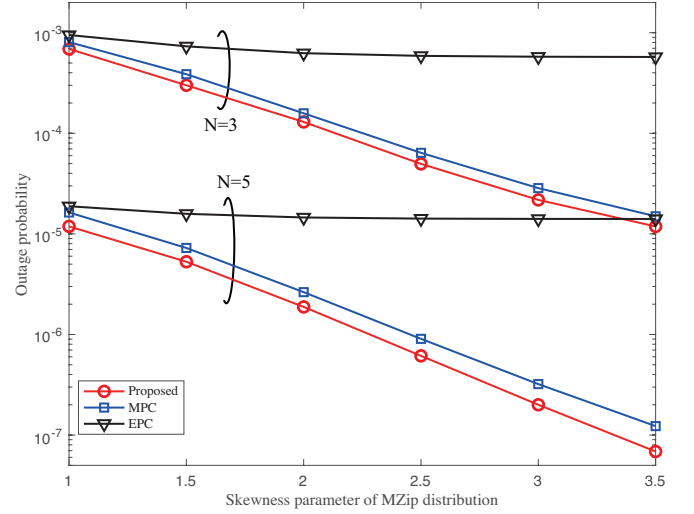


Figure 9. Outage probability of the three caching strategies versus the skewness of the Mzip distribution, where the transmit power at the relays and BS are 45dBm.

of the asymptotic outage probability vary proportional to the value of N , indicating that the proposed caching can utilize the system full diversity order by relays. In further, P_{out} decreases with a larger P_{BS} , as the increased transmit power at the BS can help effectively reduce the file delivery latency from the BS to the relays.

D. Effectiveness of the proposed caching strategy

Fig. 4 illustrates the convergence comparison between the proposed soft-BCD and the conventional BCD algorithm, where the transmit power at BS and relays are set to 45dBm. We can observe that the outage probability of the conventional BCD algorithm decreases rapidly in the first few iterations. However, the performance remains almost unchanged in the rest iterations. It is notable that the proposed soft-BCD shows superiority both in the convergence rate and the ability of

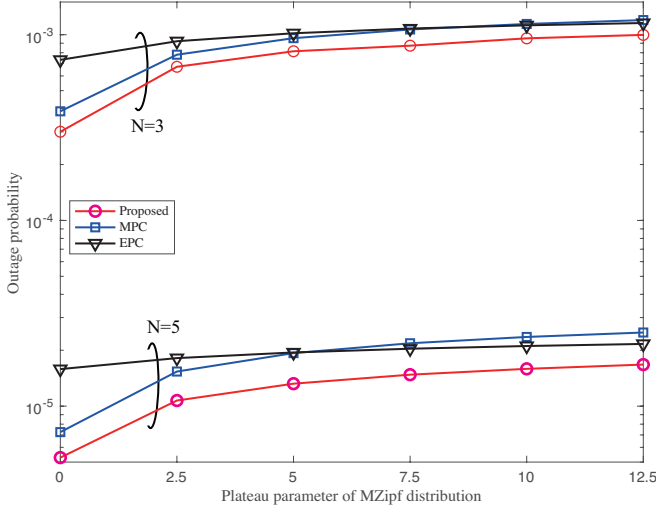


Figure 10. Outage probability of the three caching strategies versus the plateau parameter of the MZipf distribution, where the transmit power at the relays and BS are 45dBm.

global optimization. These results indicate that the soft operation can help jump over the stagnation in the iterative optimization process.

Fig. 5 compares the proposed caching with the conventional MPC and EPC caching strategies in terms of outage probability, where $N = 3$ or 5 , the transmit power at the BS P_{BS} is 45dBm and the transmit power at the relays P_R varies from 20dBm to 45dBm. From Fig. 5, we can find that the proposed caching outperforms the conventional MPC and EPC strategies, which verifies the superiority of the proposed caching due to a better utilization of both caching diversity and relaying diversity. Moreover, the outage probability of the three caching strategies becomes better with a larger P_R , as the increased transmit power at the relays can help reduce the file delivery latency from the relays to the destination. In further, the system outage probability of the three caching strategies improves with a larger N , as more relays provide a larger diversity order, which can be utilized to enhance the system transmission performance rapidly.

Fig. 6 shows the outage probability of the aforementioned three caching strategies versus the transmit power at the BS P_{BS} for different values of N , where P_R is 45dBm, the number of relays is in $\{3, 5\}$, and P_{BS} varies from 20dBm to 45dBm. From Fig. 6, we can observe that, for different values of N , the outage probability of the three caching strategies decreases with the increasing P_{BS} , as the link quality from the BS to the relays is improved. In addition, the outage probability of the three caching strategies improves along with a larger N , which shows that deploying more relays can help improve the system performance. Moreover, when $N = 3$ and $N = 5$, the proposed caching reaches the lowest outage probability among the three competing strategies. It is notable that the proposed caching with $N = 3$ even outperforms the other strategies with $N = 5$ in the low P_{BS} region. This is because in this circumstance, the link between the BS and relays becomes the bottleneck of the system, and it is hard to improve the system performance by only

deploying more relays, whereas the system performance can be further improved by the proposed caching. These results further indicate the superiority of the proposed caching over the competing strategies.

Fig. 7 shows the impact of the cache size at the relays C_1 on the outage probability of the aforementioned caching strategies, where P_{BS} and P_R are 45dBm, $N \in \{3, 5\}$, and C_1 varies from 1 to 9. From this figure, we can observe that for different values of N , the outage probability of the three caching strategies improves with the increasing C_1 . This is because the relays can pre-store more requested files, leading to faster response. Moreover, the outage probability of the three caching strategies improves with a larger N , as deploying more relays can increase the system spatial diversity. Another interesting phenomenon is that when the total cache size is equal, i.e., identical $N \times C_1$, a larger N can provide better system performance. This shows an important insight on the system design: the system performance is more sensitive to the number of relays than the cache size when the cache size is limited. This is because more relays can provide a better channel gain to access S in the case of missing caches. In further, the outage probability of the proposed caching is always lower than that of other caching strategies for different values of N , and the performance gap between the proposed caching and the MPC strategy also enlarges. This is because the proposed strategy can fully utilize the cache space to improve the system performance, which further demonstrates the superiority of the proposed caching.

Fig. 8 depicts the impact of cache size at the BS C_2 on the outage probability of the aforementioned three caching strategies, where P_R and P_{BS} are both 45dBm, N varies in $\{3, 5\}$, and C_2 varies from 5 to 25. As we can observe from the figure, the outage probability of the three caching strategies improves with the increasing C_2 for different values of N , as the BS is able to store more files, and the communication overhead between BS and S is thereby reduced, which leads to a lower latency for a fixed channel. In addition, we can still find that deploying more relays can profoundly improve the system performance. In a nutshell, the proposed caching still outperforms the other caching strategies in this case, which reveals that the proposed caching can effectively exploit the cache space at the relays and BS.

Fig. 9 shows the impact of the skewness parameter η of the MZipf distribution on the outage probability of the proposed caching, MPC and EPC schemes, where N varies in $\{3, 5\}$, P_R and P_{BS} are both set to 45dBm, and C_1 and C_2 are 3 and 10, respectively. In this figure, η varies from 1 to 3.5, where a large value of η indicates a highly concentrated file popularity, while a small one converges to uniform popularity. We can observe from this figure that the performance of the EPC scheme is almost unaffected by the increase of η , and is the worst among the three schemes since the EPC treats all files equally and ignores the impact of the multi-tier network architecture on the caching and file delivery. Moreover, the proposed caching and MPC schemes outperform the EPC, while the performance gap becomes wider with an increasing η . This is because that these two schemes can exploit the file popularity to make cache decisions. Importantly, the proposed caching scheme

achieves the best performance whatever η is, since it makes the collaborative cache decisions from the global perspective, which verifies the proposed scheme's advantages.

Fig. 10 illustrates the impact of the plateau parameter τ of the MZipf distribution on the outage probability of the proposed caching, MPC and EPC schemes, where τ varies from 0 to 12.5. With the increase of τ , the popularity distribution becomes more flattened, which means that the file popularity tends to be equal. From this figure, we can find that the performance of the three caching schemes shows an upward trend in outage probability. In particular, the curve of the EPC changes a little, because it caches the files with the same probability. Besides, the MPC is greatly affected by the increase of τ , and it performs worse than the EPC when $\tau > 7.5$. Although the proposed caching scheme is also affected by an increasing τ , it still shows the advantage over the other schemes. This is because the proposed optimization algorithm can adapt to the change of τ , and make efficient caching decisions. These results further demonstrate the superiority of the proposed caching scheme over the conventional ones.

VI. CONCLUSIONS

This work investigated a collaborative cache-aided relaying network, where the destination was randomly located in the network and it requested files from the source through the help of cache-aided BS and N relays. To analyze the system performance, we first derived the analytical P_{out} , by fully considering the random location of the destination and different file delivery modes related to the file caching status. We then performed the asymptotic analysis on the system outage probability in the high regime of SNR, to reveal some important and meaningful insights on the network. We further optimized the caching strategies among the relays and BS, to enhance the system outage probability. Simulation results were finally presented to confirm the accuracy of the derived analytical and asymptotic expressions of the proposed caching strategy. It was also shown that the proposed caching is superior to the conventional caching strategies such as MPC and EPC caching strategy.

APPENDIX A PROOF OF THEOREM 1

When $g_{1,k} \geq 1$, the outage event occurs when the latency $t_{k,1}$ and $t_{k,2}$ are both above the threshold γ_{th} . Therefore, the associated outage expression of $P_{out,1}$ can be given as

$$\begin{aligned} P_{out,1} &= \Pr(t_{k,1} > \gamma_{th}, t_{k,2} > \gamma_{th}) \\ &= \int_0^R \underbrace{\Pr\left(\log_2\left(1 + P_R d_1^{-\sigma} v_{1n_1^*}/N_o\right) < \frac{Q}{\gamma_{th}}\right)}_{\Theta_1} \\ &\quad \times \underbrace{\Pr\left(\log_2\left(1 + P_R d_1^{-\sigma} v_{1n_2^*}/N_o\right) < \frac{Q}{\gamma_{th} - T_1}\right)}_{\Theta_2} \\ &\quad \times f_{d_1}(d_1) dd_1, \end{aligned} \quad (\text{A.1})$$

where $f_{d_1}(d_1) = \frac{2d_1}{R^2}$ is the probability density function (PDF) of the distance between D and relays. Then, we derive the analytical expression of ζ_1 as,

$$\Theta_1 = \Pr\left(v_{1n_1^*} < \frac{\gamma_{1,1} N_o}{P_R d_1^{-\sigma}}\right). \quad (\text{A.2})$$

The CDF of $v_{1n_1^*}$ can be given by

$$\begin{aligned} F_{v_{1n_1^*}}(v) &= \left(1 - e^{-v}\right)^{g_{1,k}} \\ &= \sum_{n=0}^{g_{1,k}} \binom{g_{1,k}}{n} (-1)^n e^{-nv}. \end{aligned} \quad (\text{A.3})$$

By applying the above CDF into (A.2), we have Θ_1 as,

$$\Theta_1 = \sum_{n=0}^{g_{1,k}} \binom{g_{1,k}}{n} (-1)^n \exp\left(-\frac{n\gamma_{1,1} N_o}{P_R d_1^{-\sigma}}\right). \quad (\text{A.4})$$

Similarly, Θ_2 can be derived as,

$$\begin{aligned} \Theta_2 &= \Pr\left(v_{1n_2^*} < \frac{\gamma_{1,2} N_o}{P_R d_1^{-\sigma}}\right) \\ &= \sum_{n=0}^{N-g_{1,k}} \binom{N-g_{1,k}}{n} (-1)^n \exp\left(-\frac{n\gamma_{1,2} N_o}{P_R d_1^{-\sigma}}\right). \end{aligned} \quad (\text{A.5})$$

Next, substituting (A.4) and (A.5) into (A.1), we can obtain $P_{out,1}$ in (A.6). To further solve this integral, we separate it into three parts to apply the transform ϕ in [35, eq. (3.381.4)], where $\gamma(\cdot, \cdot)$ represents the lower incomplete Gamma function. In this way, we have finished the proof of Theorem 1.

APPENDIX B PROOF OF THEOREM 2

To prove Theorem 2, we write the expression of $P_{out,2}$ as,

$$\begin{aligned} P_{out,2} &= \Pr\left(\max_{n \in \mathcal{N}} \min(P_{BS} d_2^{-\sigma} v_{2n}, P_R d_1^{-\sigma} v_{1n}) < \gamma_2 N_o\right) \\ &= \int_0^R \left(1 - \exp\left(-\frac{\gamma_2 N_o}{P_R d_1^{-\sigma}} - \frac{\gamma_2 N_o}{P_{BS} d_2^{-\sigma}}\right)\right)^N f_{d_1}(d_1) dd_1. \end{aligned} \quad (\text{B.1})$$

By applying the PDF of $f_{d_1}(d_1) = \frac{2d_1}{R^2}$ into the above equation, and then solving the required integral, we can obtain $P_{out,2}$ in (B.2), which has proven Theorem 2.

APPENDIX C PROOF OF THEOREM 4

When the transmit SNR is large enough, we can use the exponential approximation of $e^x \simeq 1 + x$ to simplify the asymptotic expression of $P_{out,1}$. Specifically, Θ_1 can be approximated by

$$\Theta_1 \simeq \left(\frac{\gamma_{1,1} N_o}{P_R d_1^{-\sigma}}\right)^{g_{1,k}}. \quad (\text{C.1})$$

$$\begin{aligned}
P_{out,1}(g_{1,k}) &= \frac{2}{R^2} \int_0^R \sum_{n_1=0}^{g_{1,k}} \sum_{n_2=0}^{N-g_{1,k}} \binom{g_{1,k}}{n_1} \binom{N-g_{1,k}}{n_2} (-1)^{(n_1+n_2)} d_1 \exp\left(-\frac{(n_1\gamma_{1,1} + n_2\gamma_{1,2})N_o}{P_R d_1^{-\sigma}}\right) dd_1 \\
&= \frac{2}{R^2} \left(\int_0^R d_1 dd_1 + \int_0^R \sum_{n_1=0}^{g_{1,k}} \sum_{n_2=1}^{N-g_{1,k}} \binom{g_{1,k}}{n_1} \binom{N-g_{1,k}}{n_2} (-1)^{(n_1+n_2)} d_1 \exp\left(-\frac{(n_1\gamma_{1,1} + n_2\gamma_{1,2})N_o}{P_R d_1^{-\sigma}}\right) dd_1 \right. \\
&\quad \left. + \int_0^R \sum_{n_1=1}^{g_{1,k}} \binom{g_{1,k}}{n_1} (-1)^{(n_1)} d_1 \exp\left(-\frac{(n_1\gamma_{1,1})N_o}{P_R d_1^{-\sigma}}\right) dd_1 \right) \\
&\stackrel{(\phi)}{=} 1 + \frac{2}{R^2} \sum_{n_1=0}^{g_{1,k}} \sum_{n_2=1}^{N-g_{1,k}} \binom{g_{1,k}}{n_1} \binom{N-g_{1,k}}{n_2} (-1)^{(n_1+n_2)} \frac{\gamma(\frac{2}{\sigma}, \beta_{1,1} R^\sigma)}{\sigma \beta_{1,1}^{\frac{2}{\sigma}}} + \frac{2}{R^2} \sum_{n_1=1}^{g_{1,k}} \binom{g_{1,k}}{n_1} (-1)^{n_1} \frac{\gamma(\frac{2}{\sigma}, \beta_{1,2} R^\sigma)}{\sigma \beta_{1,2}^{\frac{2}{\sigma}}}. \tag{A.6}
\end{aligned}$$

$$\begin{aligned}
P_{out,2} &= 1 + \frac{2}{R^2} \sum_{n=1}^N \binom{N}{n} (-1)^{(n)} \exp\left(-\frac{n\gamma_2 N_o}{P_{BS} d_2^{-\sigma}}\right) \int_0^R d_1 \exp\left(-\frac{n\gamma_2 N_o}{P_R d_1^{-\sigma}}\right) dd_1 \\
&= 1 + \frac{2}{R^2} \sum_{n=1}^N \binom{N}{n} (-1)^{(n)} \exp\left(-\frac{n\gamma_2 N_o}{P_{BS} d_2^{-\sigma}}\right) \frac{\gamma(\frac{2}{\sigma}, \beta_2 R^\sigma)}{\sigma \beta_2^{\frac{2}{\sigma}}}. \tag{B.2}
\end{aligned}$$

Note that Θ_2 can be approximated as well. Hence, we can present the asymptotic expression of $P_{out,1}(g_{1,k})$ as

$$\begin{aligned}
P_{out,1}(g_{1,k}) &\simeq \frac{2}{R^2} \int_0^R \left(\frac{\gamma_{1,1} N_o}{P_R d_1^{-\sigma}}\right)^{g_{1,k}} \left(\frac{\gamma_{1,2} N_o}{P_R d_1^{-\sigma}}\right)^{N-g_{1,k}} d_1 dd_1 \\
&= \frac{2\gamma_{1,1}^{g_{1,k}} \gamma_{1,2}^{N-g_{1,k}} N_o^N}{P_R^N R^2} \int_0^R d_1^{N\sigma+1} dd_1 \\
&= \frac{2\gamma_{1,1}^{g_{1,k}} \gamma_{1,2}^{N-g_{1,k}} R^{N\sigma} N_o^N}{P_R^N (\sigma N + 2)}. \tag{C.2}
\end{aligned}$$

Similarly, we can approximate $P_{out,2}$ as

$$\begin{aligned}
P_{out,2} &\simeq \frac{2}{R^2} \int_0^R \left(\frac{\gamma_2 N_o}{P_R d_1^{-\sigma}} + \frac{\gamma_2 N_o}{P_{BS} d_2^{-\sigma}}\right)^N d_1 dd_1 \\
&= \frac{2}{R^2} \left(\frac{\gamma_2 N_o}{P_R P_{BS} d_2^{-\sigma}}\right)^N \int_0^R \left(\frac{P_R d_1^{-\sigma} + P_{BS} d_2^{-\sigma}}{d_1^{-\sigma}}\right)^N d_1 dd_1 \\
&= 2\gamma_2^N N_o^N \sum_{n=0}^N \binom{N}{n} P_R^{n-N} P_{BS}^{-n} d_2^{\sigma n} \frac{R^\delta}{\delta + 2}. \tag{C.3}
\end{aligned}$$

In a similar way, we can obtain the asymptotic expression of $P_{out,3}$, which is equivalent to that of $P_{out,2}$ through substituting γ_2 by γ_3 . In this way, Theorem 4 is proven.

APPENDIX D PROOF OF LEMMA 1

By performing the second-order derivative of P_{out} with respect to $(q_{1,1}, q_{2,1}, \dots, q_{1,K}, q_{2,K})$, we can obtain a Hessian

matrix $\mathbf{H}$ as,

$$\begin{aligned}
\mathbf{H} &= \begin{bmatrix} \frac{\partial^2 P_{out}}{\partial q_{1,1}^2} & \frac{\partial^2 P_{out}}{\partial q_{1,1} \partial q_{2,1}} & \cdots & \frac{\partial^2 P_{out}}{\partial q_{1,1} \partial q_{1,K}} & \frac{\partial^2 P_{out}}{\partial q_{1,1} \partial q_{2,K}} \\ \frac{\partial^2 P_{out}}{\partial q_{2,1} \partial q_{1,1}} & \frac{\partial^2 P_{out}}{\partial q_{2,1}^2} & \cdots & \frac{\partial^2 P_{out}}{\partial q_{2,1} \partial q_{1,K}} & \frac{\partial^2 P_{out}}{\partial q_{2,1} \partial q_{2,K}} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{\partial^2 P_{out}}{\partial q_{1,K} \partial q_{1,1}} & \frac{\partial^2 P_{out}}{\partial q_{1,K} \partial q_{2,1}} & \cdots & \frac{\partial^2 P_{out}}{\partial q_{1,K}^2} & \frac{\partial^2 P_{out}}{\partial q_{1,K} \partial q_{2,K}} \\ \frac{\partial^2 P_{out}}{\partial q_{2,K} \partial q_{1,1}} & \frac{\partial^2 P_{out}}{\partial q_{2,K} \partial q_{2,1}} & \cdots & \frac{\partial^2 P_{out}}{\partial q_{2,K} \partial q_{1,K}} & \frac{\partial^2 P_{out}}{\partial q_{2,K}^2} \end{bmatrix} \\
&= \begin{bmatrix} \mathbf{H}_{\mathcal{P}}(q_{1,1}, q_{2,1}) & & \\ & \ddots & \\ & & \mathbf{H}_{\mathcal{P}}(q_{1,K}, q_{2,K}) \end{bmatrix}, \tag{D.1}
\end{aligned}$$

where $\mathbf{H}_{\mathcal{P}}(\cdot, \cdot)$ is a sub-Hessian matrix, representing the second-order derivative of P_{out} with respect to $(q_{1,k}, q_{2,k})$. As P_{out} contains the one-nomial form of $q_{2,k}$ only, we can obtain $\frac{\partial^2 P_{out}}{\partial q_{2,k}^2} = 0$ and

$$\det \left[\mathbf{H}_{\mathcal{P}}(q_{1,k}, q_{2,k}) \right] < 0, \forall k \in [1, K]. \tag{D.2}$$

Hence, there are negative eigenvalues in $\mathbf{H}_{\mathcal{P}}$. As the eigenvalues of $\mathbf{H}$ is composed of those of $\mathbf{H}_{\mathcal{P}}(q_{1,k}, q_{2,k})$, we have finished proving that $\mathbf{H}$ is neither a definite nor semi-definite matrix.

APPENDIX E PROOF OF LEMMA 2

To prove Lemma 2, we first substitute (19), (20), and (21) into (18). For a fixed $q_{2,k}$, define

$$A_k = q_{2,k} P_{out,2} + (1 - q_{2,k}) P_{out,3}, \tag{E.1}$$

and set $p_{k,0} = A_k$ and $p_{k,g} = P_{out,1}(g)$ for $1 \leq g \leq N$. Then the k -th file contribution can be written as the Bernstein polynomial

$$\mathcal{G}_k(q_{1,k}) = \sum_{g=0}^N \binom{N}{g} p_{k,g} q_{1,k}^g (1 - q_{1,k})^{N-g}. \quad (\text{E.2})$$

Using the derivative identities of Bernstein polynomials, we have

$$\mathcal{G}'_k(q) = N \sum_{g=0}^{N-1} \binom{N-1}{g} (p_{k,g+1} - p_{k,g}) q^g (1 - q)^{N-1-g}, \quad (\text{E.3})$$

and

$$\mathcal{G}''_k(q) = N(N-1) \sum_{g=0}^{N-2} \binom{N-2}{g} \Delta^2 p_{k,g} q^g (1 - q)^{N-2-g}, \quad (\text{E.4})$$

where $\Delta^2 p_{k,g} = p_{k,g+2} - 2p_{k,g+1} + p_{k,g}$. For the considered file-delivery model, the outage sequence $\{p_{k,g}\}_{g=0}^N$ is decreasing and convex in the discrete sense, i.e., $p_{k,g+1} \leq p_{k,g}$ and $\Delta^2 p_{k,g} \geq 0$. Therefore, $\mathcal{G}'_k(q) \leq 0$ and $\mathcal{G}''_k(q) \geq 0$ for $q \in [0, 1]$. Since $P_{out} = \sum_{k=1}^K a_k \mathcal{G}_k(q_{1,k})$ with $a_k \geq 0$, P_{out} is convex and non-increasing with respect to $q_{1,k}$. This completes the proof.

APPENDIX F PROOF OF THEOREM 5

From (42), the Lagrangian of **P1-1** can be written as

$$\begin{aligned} \mathcal{L} = & \sum_{k=1}^K a_k G_k(q_{1,k}) + \lambda \left(\sum_{k=1}^K q_{1,k} - C_1 \right) \\ & + \sum_{k=1}^K (u_k(q_{1,k} - 1) - v_k q_{1,k}). \end{aligned} \quad (\text{F.1})$$

The stationarity condition is

$$a_k G'_k(q_{1,k}^*) + \lambda^* + u_k^* - v_k^* = 0. \quad (\text{F.2})$$

If $0 < q_{1,k}^* < 1$, then $u_k^* = v_k^* = 0$, and hence

$$a_k G'_k(q_{1,k}^*) + \lambda^* = 0. \quad (\text{F.3})$$

If $q_{1,k}^* = 0$, then $u_k^* = 0$ and $v_k^* = a_k G'_k(0) + \lambda^* \geq 0$, which gives

$$a_k G'_k(0) + \lambda^* \geq 0. \quad (\text{F.4})$$

If $q_{1,k}^* = 1$, then $v_k^* = 0$ and $u_k^* = -(a_k G'_k(1) + \lambda^*) \geq 0$, which gives

$$a_k G'_k(1) + \lambda^* \leq 0. \quad (\text{F.5})$$

Combining the above three cases with $\sum_{k=1}^K q_{1,k}^* = C_1$ gives (43)–(44). This completes the proof.

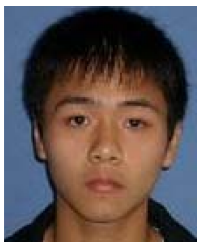
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